UWB LOS/NLOS Identification Using Spatio-Temporal Deep Learning

Liangang Qi, Yun Zhao, Qiang Guo, and Mykola Kaliuzhnyi

***Abstract*— Reliable identification of line-of-sight (LOS) and non-line-of-sight (NLOS) conditions is critical for accurate localization in ultra-wideband (UWB) systems. In this letter, we propose a Voronoi+CNN-BiLSTM architecture for LOS/NLOS classification of UWB signals. By integrating Voronoi-based spatial propagation features, the model captures geometric characteristics of the environment to enhance classification performance. Experimental results on the publicly available UPT dataset demonstrate that the proposed method achieves an overall accuracy of 97.43%. Cross-scene evaluations demonstrate robust generalization. Moreover, the model achieves an average inference time of 2.08 ms per sample on a standard CPU, making it suitable for real-time applications.**

***Index Terms*— UWB, NLOS identification, CNN, BiLSTM, Voronoi tessellation.**

I. Introduction

INDOOR localization demands high-accuracy range estimation. Ultra-wideband (UWB) technology is well-suited for this task due to its centimeter-level ranging precision, strong multipath robustness, and low power consumption. However, UWB performance degrades severely under non-line-of-sight (NLOS) conditions [1]. Obstacles between transmitter and receiver cause signals to diffract, reflect, or penetrate along indirect paths. These indirect paths delay and attenuate the received signal, introducing positive biases in time-of-flight (TOF) distance estimates [2]. Since most standard ranging algorithms assume ideal line-of-sight (LOS) propagation, unmitigated NLOS channels lead to large positioning errors [3]. Detecting and classifying LOS versus NLOS channels is therefore critical for maintaining UWB localization accuracy.

To address the challenge of NLOS identification in UWB systems, a wide range of techniques have been proposed. Early machine learning and deep learning approaches, such as Zeng et al. [4], transformed Channel Impulse Response (CIR) signals into image-like representations for CNN-based processing, which introduced high computational costs. To reduce this overhead, subsequent research shifted toward directly exploiting raw CIR time series. To address these issues, Jiang et al. in [5] proposed a CNN-LSTM architecture that directly processes raw CIR sequences, achieving higher classification accuracy without relying on handcrafted radio parameters. Transfer learning has also been explored to enhance model generalization under data-scarce conditions [6]. Building on this, Pei et al. [7] integrated self-attention modules into a fully convolutional network, achieving 88.24% accuracy on a public dataset. In parallel, Wu et al. [8] proposed a self-attention-assisted TinyML framework that attained 93.14% accuracy on a public dataset while enabling efficient NLOS classification on resource-constrained edge devices.

Beyond attention-based designs, geometry-inspired approaches have also been explored. For example, Wang et al. [9] proposed a cooperative localization algorithm for mixed LOS/NLOS environments, leveraging the geometric relationships of distance biases to mitigate NLOS effects. While these studies highlight the benefits of incorporating spatial information, most existing deep learning methods still focus primarily on temporal features extracted from CIR sequences and often neglect the broader spatial context in which UWB signals propagate. Since the physical environment—such as anchor placement and surrounding geometry—significantly influences signal behavior, especially under NLOS conditions, it is important to integrate spatial prior knowledge into the learning process. Motivated by this, we incorporate geometric features derived from anchor locations [10], enabling our model to jointly exploit temporal dynamics and environmental geometry for more robust LOS/NLOS classification.

In this study, we extract propagation-relevant points from each CIR, focusing on the significant high-energy peaks, and treat them as pseudo-anchor points that represent dominant reflectors or scattering centers. A Voronoi diagram is then constructed based on the spatial layout of these points, introducing scene-dependent geometric priors. The resulting diagram partitions the space into sub-regions reflecting the local influence of each pseudo-anchor on signal propagation. This spatial structure is rasterized into a 2D grayscale image and used as an auxiliary input to the network. By jointly learning temporal dynamics and environmental geometry, the model achieves improved classification accuracy and generalization. The main contributions of this letter are summarized as follows:

- We introduce a Voronoi-based feature encoding of anchor geometry as a spatial prior for UWB signal analysis. Each anchor's Voronoi cell is converted into a spatial mask, capturing the region of its dominant signal propagation.
- We design a hybrid deep network that jointly processes the spatial Voronoi features and the temporal UWB CIR measurements. Convolutional layers learn local patterns in the augmented input, while Bidirectional LSTM (BiLSTM) models the sequential channel evolution.

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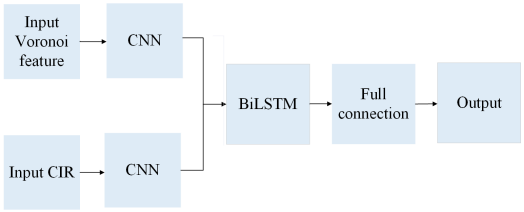


Fig. 1. Proposed neural network architecture.

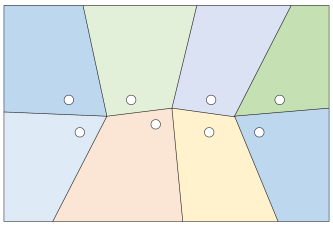


Fig. 2. Spatial partitioning based on Voronoi Tessellation, where circles represent dominant reflectors.

The remainder of the letter is organized as follows. Section II details the proposed feature construction and network architecture. Section III presents experimental results and analysis. Finally, Section IV concludes the letter.

II. Proposed System

This section outlines the architecture of the proposed deep learning-based classification framework. The method integrates multimodal feature extraction with temporal sequence modeling to comprehensively exploit both the physical propagation characteristics of UWB signals and the geometric priors of the surrounding environment. By jointly leveraging these complementary aspects, the system aims to enhance the reliability and accuracy of NLOS identification.

A. Overview

To fully exploit the spatiotemporal characteristics embedded in UWB signals, we propose a deep learning-based approach for NLOS identification. The conceptual framework and overall processing pipeline are illustrated in Fig. 1. The method begins with temporal modeling of CIR sequences, complemented by geometric feature extraction based on Voronoi tessellation, aiming to capture both signal dynamics and environmental structure. Specifically, key reflection points are first extracted from the CIR using a peak detection algorithm. A Voronoi diagram is then constructed using these key reflections, serving as a spatial representation of the signal propagation. Subsequently, both the CIR sequence and the Voronoi-based geometric features are fed into a CNN-BiLSTM network for classification between LOS and NLOS conditions. The model consists of two parallel feature extraction branches: one uses a convolutional network to process the temporal CIR sequence, while the other encodes spatial features from the Voronoi diagram. The outputs of these two branches are fused and passed into a BiLSTM module to model latent temporal dependencies, culminating in a binary classification output.

B. Voronoi Tessellation Module

In UWB signal propagation, multiple energy peaks observed in the CIR sequence typically correspond to environmental reflectors, occluders, or scattering centers. The temporal positions and amplitudes of these peaks implicitly describe the physical layout of the environment and the paths traversed by the signal. Thus, the spatial distribution of these key reflections can provide valuable geometric priors for classifying LOS and NLOS conditions.

To capture the spatial organization of these propagation paths, we employ Voronoi tessellation, a method widely used in meteorology and geographic analysis for its ability to effectively partition space and model proximity-based relationships. Specifically, we first detect a set of prominent energy peaks from each CIR sample using a combination of amplitude thresholding A_{th} and positional constraints. In our implementation, A_{th} is empirically set to 0.1, which achieves a good balance between suppressing noise-induced spurious peaks and retaining the dominant multipath components. Although adaptive thresholding strategies were also explored, we found that they significantly increased computational overhead while yielding limited performance gains. Therefore, a fixed threshold was adopted to ensure both efficiency and robustness in practice. Let the extracted set of propagation-relevant points be denoted as $\mathcal{P} = \{p_1, p_2, \ldots, p_N\}$, where each point $p_k \in \mathbb{R}^2$ represents the estimated spatial location of a multipath component.

These points are treated as propagation-relevant points in a 2D plane, upon which a Voronoi diagram is constructed. For any given location $x \in \mathbb{R}^2$, it belongs to the Voronoi region $\mathcal{R}_i$ associated with site p_i if and only if:

$$x \in \mathcal{R}_i \iff \|x - p_i\| \leq \|x - p_j\|, \quad \forall j \neq i. \tag{1}$$

The resulting Voronoi diagram divides the space into a set of convex polygonal regions, each representing the influence zone of a corresponding signal reflection point. These regions reflect the relative dominance and spatial density of multipath components, offering a geometric interpretation of the signal's propagation behavior. Compared to conventional time-domain modeling approaches, this spatial encoding provides additional structural priors that assist the model in distinguishing complex NLOS conditions.

To extract discriminative features from the Voronoi diagram, we render it as a grayscale image with fixed resolution. Each pixel value indicates either the Voronoi region it belongs to or a normalized distance-based weight from the corresponding generator point. This image is fed into a 2D convolutional neural network (2D CNN), which extracts local boundary patterns, geometric structures, and density variations through hierarchical convolution layers. The resulting feature map is encoded into a compact vector that matches the dimensionality of the CIR temporal branch.

In the next stage, this spatial feature vector is fused with the temporal features extracted from the CIR branch via a BiLSTM network. This fusion enables the model to jointly learn spatiotemporal patterns, thereby improving the robustness and accuracy of LOS/NLOS classification. As shown in Fig. 2, this module transforms CIR-derived reflection points into a Voronoi-based geometric representation, which serves as a complementary input to the deep learning architecture, enhancing its ability to understand signal propagation under various environmental conditions.

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Although UWB CIR data is one-dimensional in the time domain, its amplitude peaks reflect different propagation paths. We heuristically approximate the distribution of these paths by projecting significant energy peaks onto a 2D plane, with the horizontal axis as delay index and the vertical axis as normalized amplitude. This projection captures signal patterns distinguishing NLOS and LOS scenarios, not physical space geometry. The resulting Voronoi tessellation provides a learnable spatial encoding of multipath component distribution. This pseudo-spatial prior enables CNNs to exploit topological patterns in a grid, enhancing LOS/NLOS classification. Future work will explore theoretical links to physical space.

This 2D representation is not meant to reflect actual spatial coordinates, but rather introduces a pseudo-spatial prior that enables convolutional kernels to exploit topological patterns of peak distributions in a regular grid. While peak detection captures energy-dominant components in the CIR, it does not explicitly encode the spatial organization or relationships between them. By constructing a Voronoi diagram over selected peaks, we transform the discrete temporal features into a structured spatial representation. This Voronoi partition provides an explicit topological representation of the multipath component distribution—information that standard CNN-BiLSTM layers, which mainly model local temporal patterns, are not well suited to capture directly from raw CIR sequences. This allows CNNs to exploit geometric patterns and spatial correlations among multipath components, enhancing the model's ability to distinguish between LOS and NLOS scenarios.

Fig. 3 provides a visual example of the Voronoi feature construction process. The upper subplot shows the normalized CIR amplitude with the detected local peaks marked in red. These high-energy components are treated as anchor points. The lower subplot displays the resulting Voronoi diagram generated from these points in the 2D time-amplitude space. Each cell in the diagram represents the spatial influence region of a corresponding peak. By encoding this structure into a binary image, the model is able to capture the geometric layout of multipath components, offering additional spatial priors for improving classification performance.

C. CNN

In this letter, CNNs are employed to encode two types of input features: one-dimensional CIR sequences and two-dimensional Voronoi diagrams. The CNN modules are designed to extract discriminative local patterns from the inputs and transform them into fixed-length high-dimensional feature vectors for subsequent temporal modeling. For the CIR input, the signal is first treated as a one-dimensional time series and processed by a three-layer 1D convolutional network. Each convolutional layer captures local amplitude variations within temporal windows, enhancing the model's ability to perceive multipath delays, reflection characteristics, and energy distributions. After each convolutional operation, ReLU activation and 1D max-pooling are applied to reduce temporal resolution while preserving key features. The resulting feature map is then flattened and passed through a fully connected layer to produce a compact representation vector.

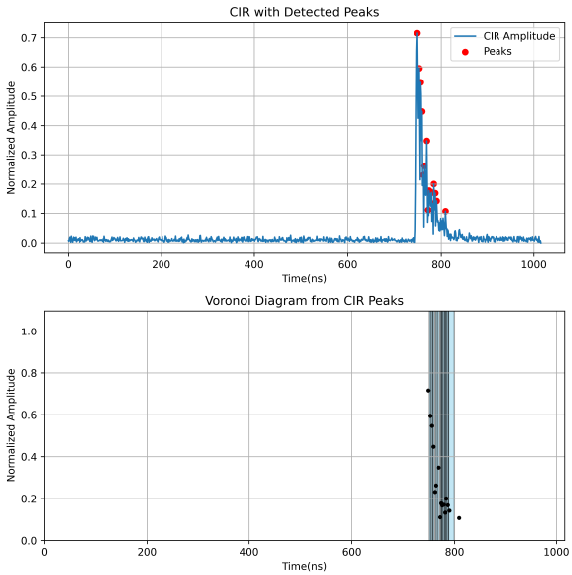


Fig. 3. Voronoi-based Representation of CIR Peaks.

For the Voronoi diagrams, a 2D CNN is employed to encode the spatial features. This branch also consists of three convolutional layers, each followed by ReLU activation and 2D max-pooling. The network progressively captures geometric attributes such as region boundaries, polygon shapes, and spatial density distributions. The final feature map is flattened into a vector, dimensionally aligned with the CIR representation, enabling effective feature fusion.

This dual-branch CNN architecture allows the model to concurrently extract features from both temporal and spatial domains. While the CIR branch focuses on temporal propagation behaviors, the Voronoi branch captures geometric cues reflecting the environment layout. The outputs from both branches are concatenated and fed into a BiLSTM for sequence modeling and classification.

D. BiLSTM

To effectively capture temporal patterns in CIR evolution, we adopt a BiLSTM architecture. Unlike standard RNNs, LSTM mitigates gradient vanishing through gated memory units, enabling long-term dependency modeling. The bidirectional structure further allows the network to exploit both past and future context, which is particularly beneficial for characterizing multipath delay profiles and reflection persistence. The final hidden states are passed through a fully connected layer to perform binary classification. This structure provides enhanced temporal context awareness and improves classification robustness under NLOS scenarios.

III. Dataset and Experimental Results

A. Dataset Description

The proposed method is evaluated using the publicly available UPT dataset [11], which includes both LOS and NLOS measurements collected across four distinct indoor environments: residential apartment, small apartment, industrial workshop, and office environment. A total of 52,000 samples were randomly selected from the dataset, with an

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TABLE I
PARAMETER INFORMATION OF THE PROPOSED MODEL

Parameter tensor	Size	Params	Parameter tensor	Size	Params
conv1_cir.weight	[16,1,3]	48	conv1_cir.bias	[16]	16
conv2_cir.weight	[32,16,3]	1536	conv2_cir.bias	[32]	32
conv3_cir.weight	[64,32,3]	6144	conv3_cir.bias	[64]	64
fc_cir.weight	[128,8128]	1040384	fc_cir.bias	[128]	128
vor_conv1.weight	[16,1,3,3]	144	vor_conv1.bias	[16]	16
vor_conv2.weight	[32,16,3,3]	4608	vor_conv2.bias	[32]	32
vor_conv3.weight	[64,32,3,3]	18432	vor_conv3.bias	[64]	64
vor_fc.weight	[128,1024]	131072	vor_fc.bias	[128]	128
lstm.weight_ih_l0	[512,256]	131072	lstm.weight_hh_l0	[512,128]	65536
lstm.bias_ih_l0	[512]	512	lstm.bias_hh_l0	[512]	512
lstm.weight_ih_l0_rev	[512,256]	131072	lstm.weight_hh_l0_rev	[512,128]	65536
lstm.bias_ih_l0_rev	[512]	512	lstm.bias_hh_l0_rev	[512]	512
lstm.weight_ih_l1	[512,256]	131072	lstm.weight_hh_l1	[512,128]	65536
lstm.bias_ih_l1	[512]	512	lstm.bias_hh_l1	[512]	512
lstm.weight_ih_l1_rev	[512,256]	131072	lstm.weight_hh_l1_rev	[512,128]	65536
lstm.bias_ih_l1_rev	[512]	512	lstm.bias_hh_l1_rev	[512]	512
fc_lstm.weight	[2,256]	512	fc_lstm.bias	[2]	2

equal number representing LOS and NLOS signal conditions. These samples were shuffled and partitioned into three subsets: 70% for training, 10% for validation, and 20% for testing. The validation set was used to monitor model performance during training and to mitigate overfitting. To enhance generalization and reduce the likelihood of over-reliance on environment-specific features, randomization was applied throughout the sample allocation process.

B. Experimental Setting Up

Parameter details are summarized in Table I. The temporal branch processes 1D CIR sequences using three convolutional layers whose trainable parameters are listed as conv1_cir.weight, conv2_cir.weight, and conv3_cir.weight, followed by a fully connected projection layer fc_cir.weight that outputs a 128-dimensional feature vector.

Similarly, the spatial branch extracts features from Voronoi-encoded grayscale images through three 2D convolutional layers, represented by vor_conv1.weight, vor_conv2.weight, and vor_conv3.weight, and then projected via vor_fc.weight to another 128-dimensional representation.

The concatenated features from both branches are passed into a BiLSTM network. Its input-to-hidden and hidden-to-hidden weights for each layer are respectively labeled as lstm.weight_ih_l0, lstm.weight_hh_l0, lstm.weight_ih_l1, and lstm.weight_hh_l1, accompanied by their corresponding biases.

Finally, the fused output is mapped to the binary classification space using fc_lstm.weight and fc_lstm.bias. This modular design allows efficient modeling of both spatial and temporal dependencies for accurate LOS/NLOS discrimination.

The proposed method was implemented in Python 3.12.3 using the PyTorch 2.4.1 framework. All experiments were conducted on a laptop equipped with an NVIDIA GeForce RTX 4060 GPU and CUDA 12.4. The model was trained with a learning rate of 0.001 and a batch size of 32. To evaluate the method's efficiency in resource-limited scenarios, additional experiments were also performed on the CPU, demonstrating that the proposed model maintains competitive performance even without GPU acceleration.

C. Evaluations on the Test Set

Table II compares the classification performance of different models on the test set. While CNN-LSTM, CNN-BiLSTM, and ADCNN-BiLSTM-Transformer achieve accuracies of 84.61%, 90.90%, and 94.64%, respectively, the proposed Voronoi+CNN-BiLSTM framework attains 97.43% accuracy, representing a 2.79 percentage point improvement over ADCNN-BiLSTM-Transformer, demonstrating the effectiveness of integrating Voronoi-based spatial features with temporal modeling.

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE TEST SET

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
CNN-LSTM	84.61	86.39	84.61	84.41
CNN-BiLSTM [6]	90.90	90.94	90.90	90.89
ADCNN-BiLSTM-Transformer [12]	94.64	94.78	94.64	94.63
Voronoi+CNN-BiLSTM	97.43	98.08	96.74	97.41

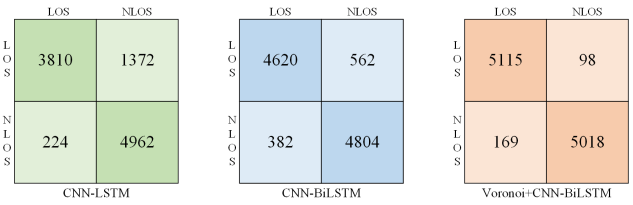


Fig. 4. Confusion matrix.

TABLE III
MODEL PERFORMANCE ON UNSEEN INDIVIDUAL ENVIRONMENTS FOR GENERALIZATION EVALUATION

Environment	Type	Precision	Recall	F1-score	Accuracy
Residential Apartment	LOS	0.70	0.93	0.80	0.83
	NLOS	0.96	0.78	0.86	
Small Apartment	LOS	0.89	0.91	0.90	0.85
	NLOS	0.75	0.72	0.74	
Office Environment	LOS	0.92	0.56	0.70	0.82
	NLOS	0.80	0.97	0.87	

Confusion matrix analysis is a widely used method for evaluating classification performance, as it provides detailed insights into both prediction accuracy and error distribution. In Fig. 4, the rows represent the ground truth labels, while the columns indicate the predicted labels. The confusion matrix illustrates the differences in classification performance across models. Compared with CNN-LSTM and CNN-BiLSTM, the proposed Voronoi+CNN-BiLSTM model achieves more balanced predictions with significantly fewer misclassifications, demonstrating its effectiveness in capturing both spatial and temporal features of UWB signal propagation.

To evaluate the model's generalization capability across diverse indoor environments, we performed an environment-exclusive test in which one location was reserved as the test set, while the remaining three were used for training and validation. As shown in Table III, the model maintained relatively high accuracy in the residential apartment (83%), small apartment (85%), and office environment (82%), demonstrating good generalization in typical indoor scenarios.

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TABLE IV
IMPACT OF INCLUDING PARTIAL INDUSTRIAL WORKSHOP DATA ON MODEL PERFORMANCE

Configuration	Type	Precision	Recall	F1-score	Accuracy
Industrial Workshop	LOS	0.91	0.23	0.37	0.55
	NLOS	0.50	0.97	0.66	
Industrial Workshop +10%	LOS	0.88	0.61	0.72	0.74
	NLOS	0.66	0.90	0.76	
Industrial Workshop +20%	LOS	0.83	0.76	0.79	0.79
	NLOS	0.75	0.83	0.79	
Industrial Workshop +30%	LOS	0.81	0.86	0.83	0.81
	NLOS	0.81	0.74	0.77	

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT METHODS

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
CNN-BiLSTM	90.90	90.94	90.90	90.89
Voronoi+CNN-BiLSTM	97.43	98.08	96.74	97.41

However, when evaluated on the industrial workshop environment, the model's performance dropped significantly, with accuracy declining to 55%. This sharp degradation can be attributed to the unique characteristics of the workshop, which features a large open space densely populated with machinery and equipment. The outer walls are made of brick and partially covered with glass, while the interior is dominated by metallic structures, resulting in highly reflective surfaces that cause severe multipath propagation and signal distortion.

To address this issue, we incrementally included 10%, 20%, and 30% of the industrial workshop data into the training set to improve domain adaptation. As summarized in Table IV, even a small fraction of in-domain data led to substantial improvements. With just 10% of the workshop data added, accuracy rose to 74%, and continued to improve to 79% and 81% with 20% and 30% inclusion, respectively. The F1-scores for both LOS and NLOS signals also improved consistently, indicating enhanced discrimination ability in the complex workshop environment. These results highlight not only the model's sensitivity to domain shift but also its strong capacity for fast adaptation when exposed to limited representative samples from new environments.

We evaluated the inference efficiency of the proposed Voronoi+CNN-BiLSTM model on both CPU and GPU platforms (52000 samples, NVIDIA GeForce RTX 4060 GPU). On a standard CPU, the average inference time per sample is 2.08 ms, including all submodules (CIR CNN, Voronoi CNN, and BiLSTM). For GPU execution, the average batch inference time is 1.531 ms (standard deviation 0.658 ms), or 0.048 ms per sample (batch size of 32). Voronoi feature generation is a one-time cost of 1.39 ms per sample, which is significantly reduced to 0.002 ms per sample through caching. As shown in Table II, the model significantly outperforms baselines, justifying the computational cost and supporting real-time UWB-based indoor applications.

D. Ablation Studies

An ablation experiment was performed to evaluate the contribution of the Voronoi feature. As reported in Table V, adding Voronoi information to the CNN-BiLSTM model consistently boosts performance, raising accuracy from 90.90% to 97.43% and F1-score from 90.89% to 97.41%. This demonstrates that the Voronoi representation effectively enhances the model's classification performance.

IV. CONCLUSION AND FUTURE PERSPECTIVES

In this letter, we proposed a Voronoi+CNN-BiLSTM architecture for LOS/NLOS classification of UWB signals. By integrating Voronoi-based spatial propagation features, the model effectively captures the geometric characteristics of the environment, resulting in improved classification performance. Experiments on a public UWB dataset demonstrate the effectiveness of the proposed method. Cross-scene evaluations further validate the robustness and generalization capability of the model. In addition, the model achieves low inference latency on a standard CPU, supporting its suitability for real-time and resource-constrained applications. Future work will focus on lightweight network designs and domain adaptation strategies to enhance adaptability in unseen deployment scenarios.

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